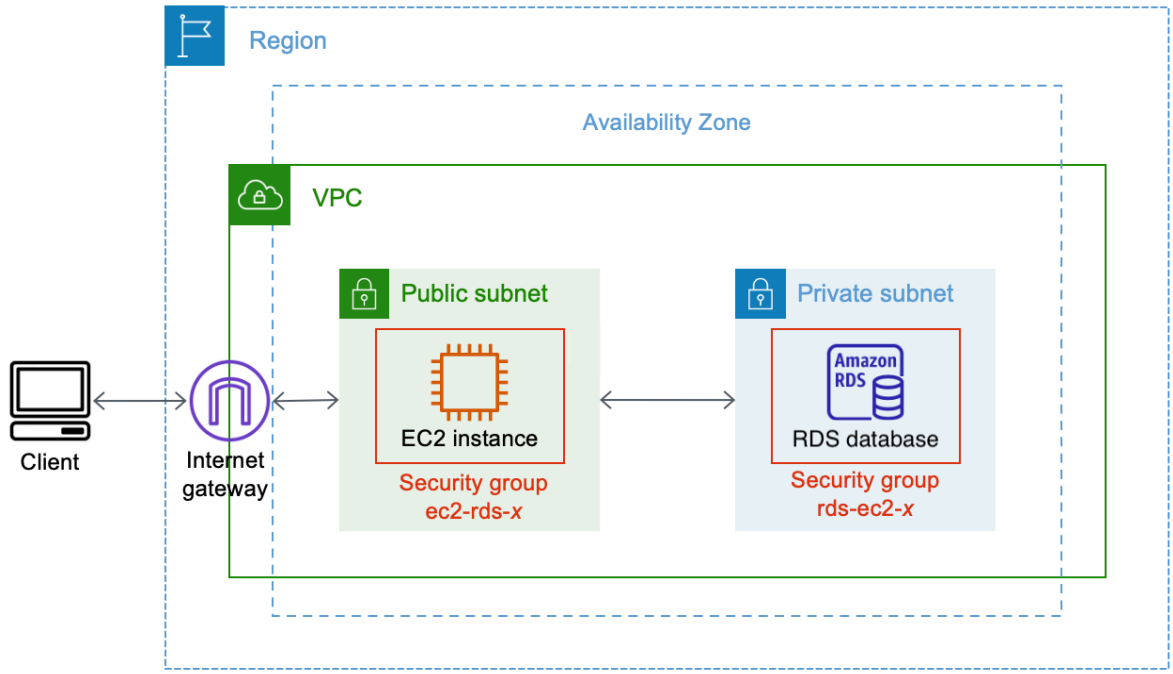
PRÁCTICA MÓDULO 2 - PRIMERA PARTE

En esta práctica, debe crear su propio servidor web:



EC2: subnet-00227b47a5f26ec04 public EC2 EC2-GrupodeSeguridad-AWS ID sg-0af17416a233137b2

RDS: subnet-056e633061b55dcbe private RDS sg-0abdd59dea4acc995

Realice el exámen en el siguiente orden:

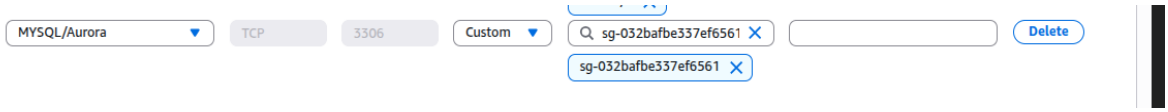
1. Crear un par de claves nuevo para poder conectarse a su servidor vía SSH
2. Crear dos grupos de seguridad uno para la EC2 y otro para la RDS
3. Lanzar una instancia de servidor web **Amazon EC2 - t2.micro 8GB de almacenamiento , solo escoger el de capa gratuita**
4. Lanzar una RDS **Amazon RDS - db.t3.micro 5GB de almacenamiento tipo mysql no aurora mysql, solo escoger el de capa gratuita**, configurar usuario y contraseña para conectarse.

PASOS:

1. Crear un par de claves (keypair) nuevo .pem

   Indispensable para poder conectarse a la instancia

2. Crear los grupos de seguridad :
   - El grupo de seguridad del EC2 debe tener permitido el puerto 22 desde tu instancia para hacer SSH
   - El grupo de seguridad que vas a usar en la RDS debe permitir el tráfico por 3306 o mysql al grupo de seguridad de la instancia, el de la imagen es solo un ejemplo pero le estas permitiendo el trafico a el grupo de la EC2 para que pueda entrar el SG de la RDS.



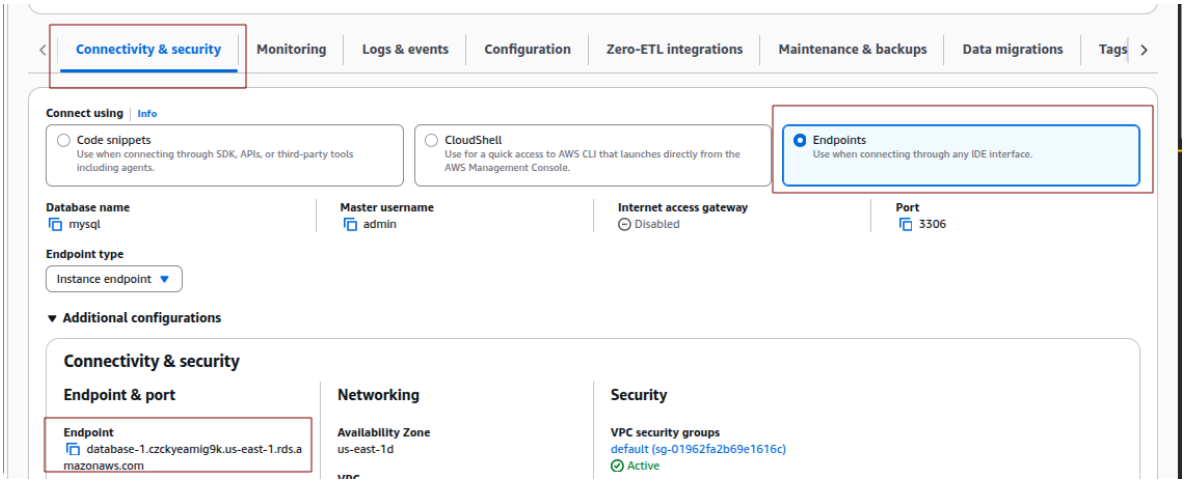
3. Lanzar la RDS: RECUERDA **anotar el usuario y contraseña** en algún lugar, tiene que ser **MySQL NO AuroraMySQL**, adicional tiene que ser en la **capa gratuita**.

4. Lanzar la EC2, permitir el tráfico SSH desde tu dirección IP.

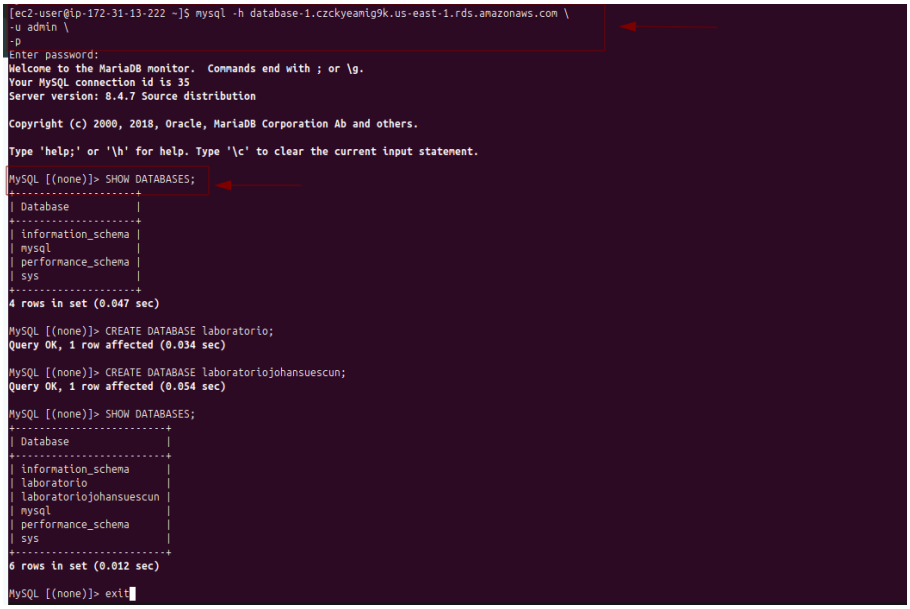
5. Conectate a la EC2, instala el MariaDB
   - Si es Amazon Linux con el comando **sudo dnf install mariadb105**

mysql -h database-aws-example.czckyeamig9k.us-east-1.rds.amazonaws.com

6. Posteriormente conectate al la RDS desde la EC2 con el comando:
   - **mysql -h database-1.czckyeamig9k.us-east-1.rds.amazonaws.com \ -u admin \ -p**
   - El endpoint lo encontrarás en tu RDS y el usuario fue el que se configuró al momento de lanzar la RDS junto con la contraseña que se le asignó al momento de lanzarla



7. Ya conectado a la RDS, crear una Base de datos:
    - **CREATE DATABASE laboratorio<u>tunombre</u>;** para crear una base de datos que se llame laboratorio
    - SHOW DATABASES; Mostrar las bases de datos creadas
    - Para asegurar que se conectó correctamente su consola se deberá ver así:



8. Subir evidencia
    - Print Screen de resumen de configuración de lanzamiento de EC2
    - Print screen RDS y el grupo de seguridad
    - Print Screen donde se vea conectado a la RDS desde la EC2

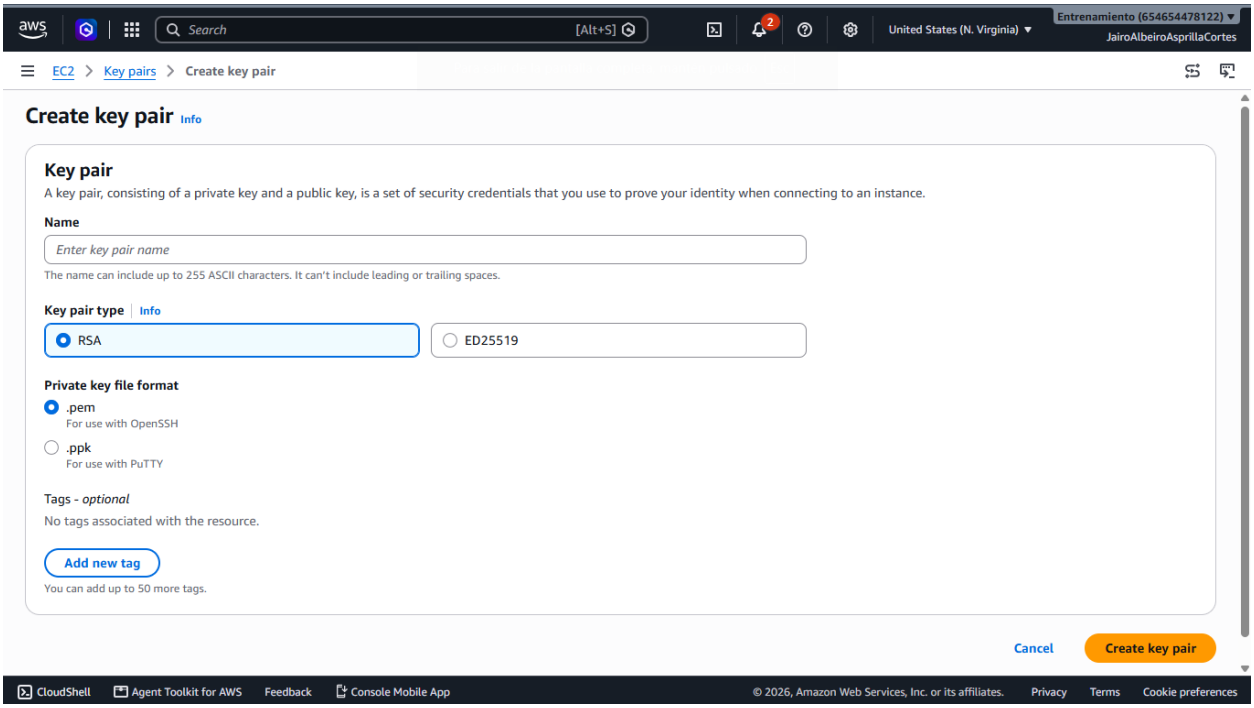
- Print screen donde se vea comando de crear database y showdatabase donde muestre la base de datos con su nombre
- Subirlo en un solo archivo ya sea PDF nombrarlo <Su_Nombre>-Modulo2-Examen1.

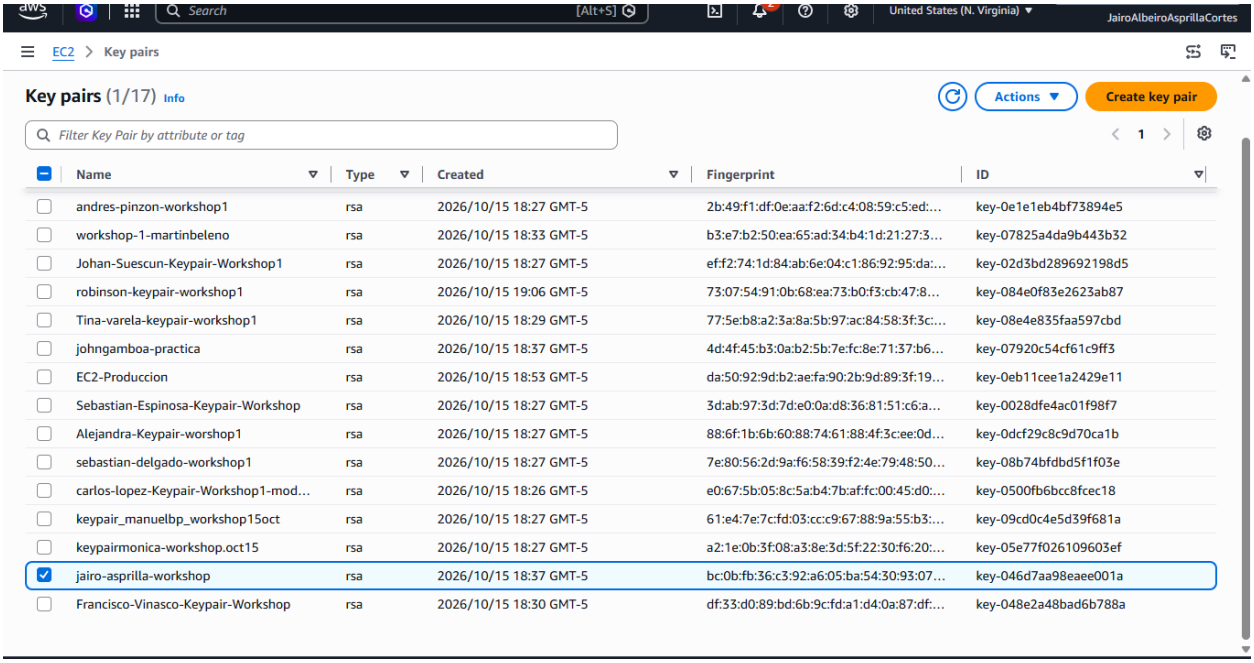
  ejemplo: PepitoPerez-Modulo2-ExamenPractico1

RECUERDA: Siempre eliminar todos los recursos y servicios desplegados, luego de tomar las respectivas evidencias

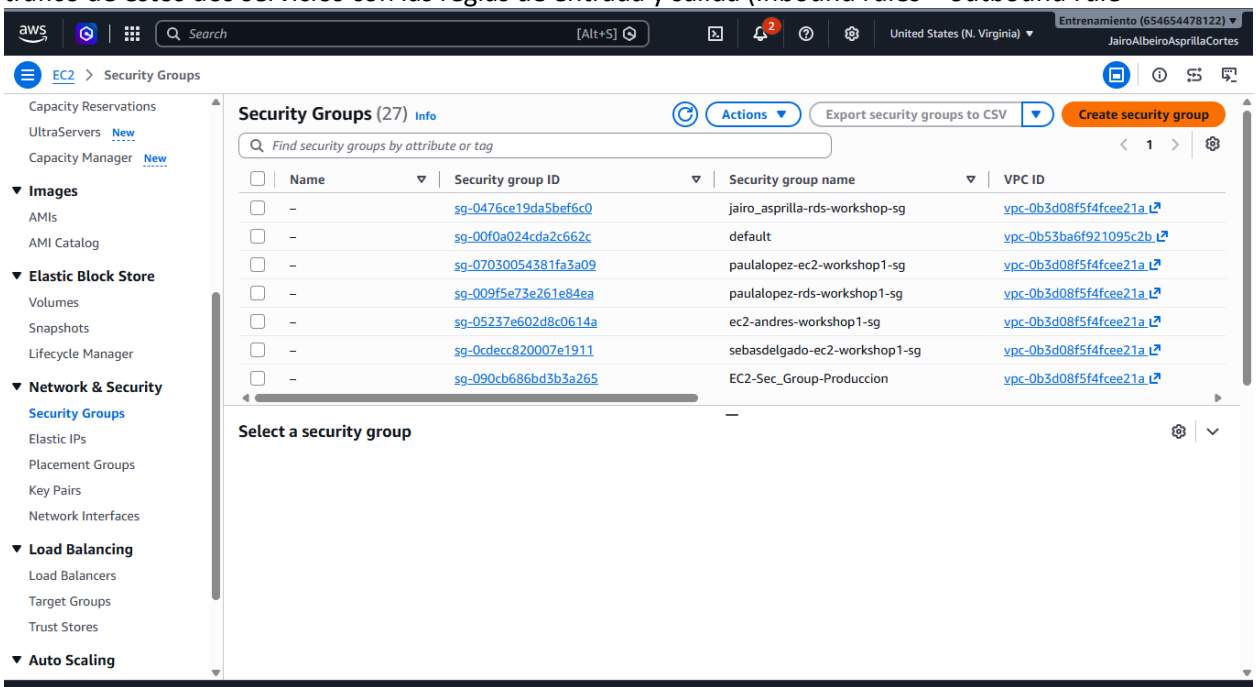
DESARROLLO

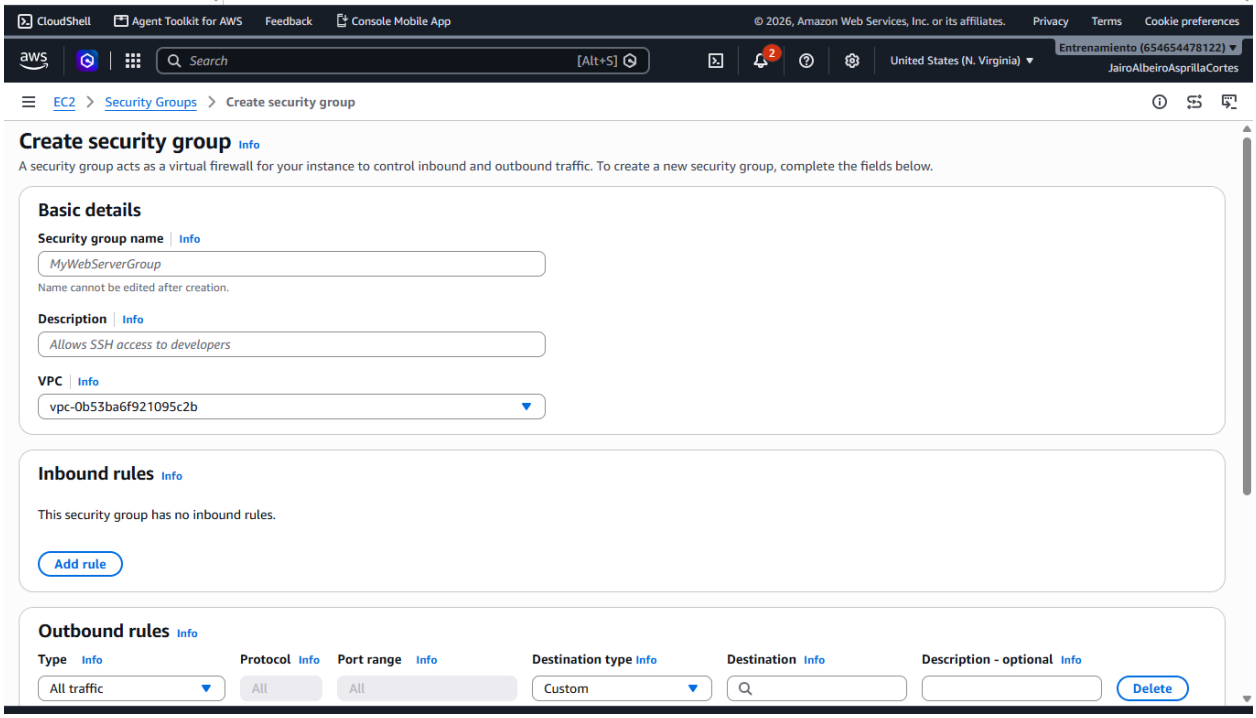
1- Creación de la kay pair para ingreso SSH a nuestra instancia EC2

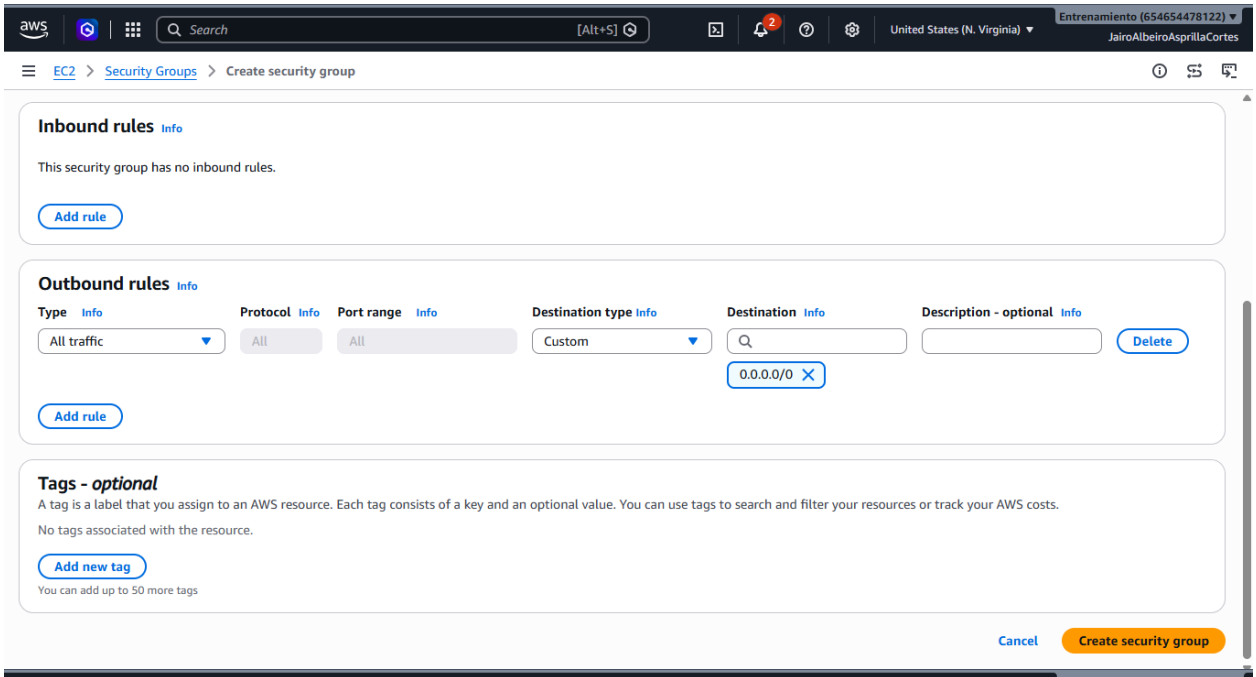


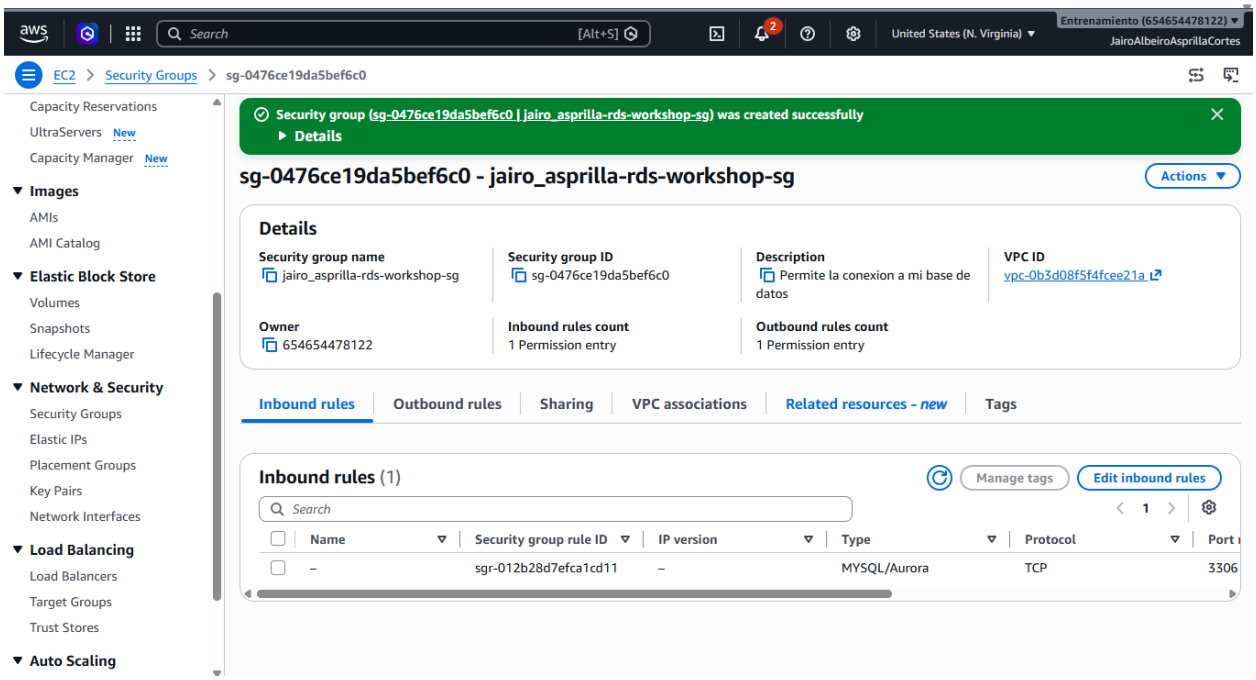


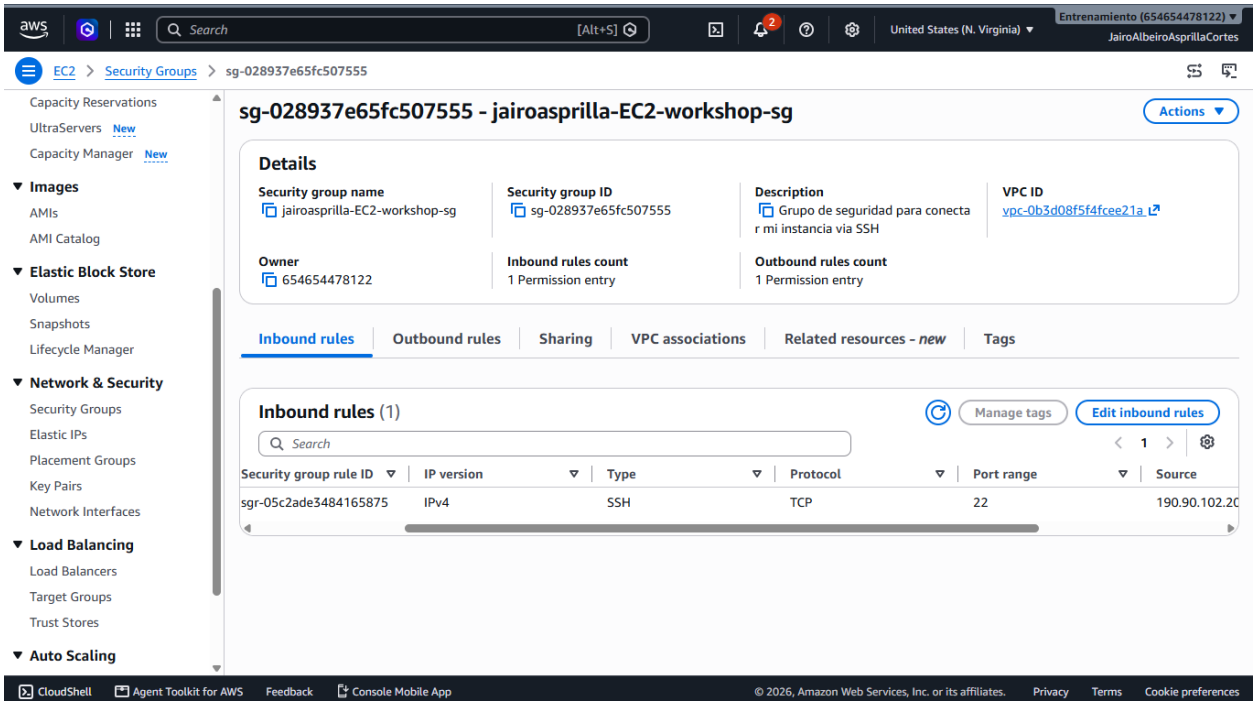
2- Creación de los grupos de seguridad para la instancia EC2 y la RDS MySQL para poder controlar el trafico de estos dos servicios con las reglas de entrada y salida (Inbound rules – outbound rule



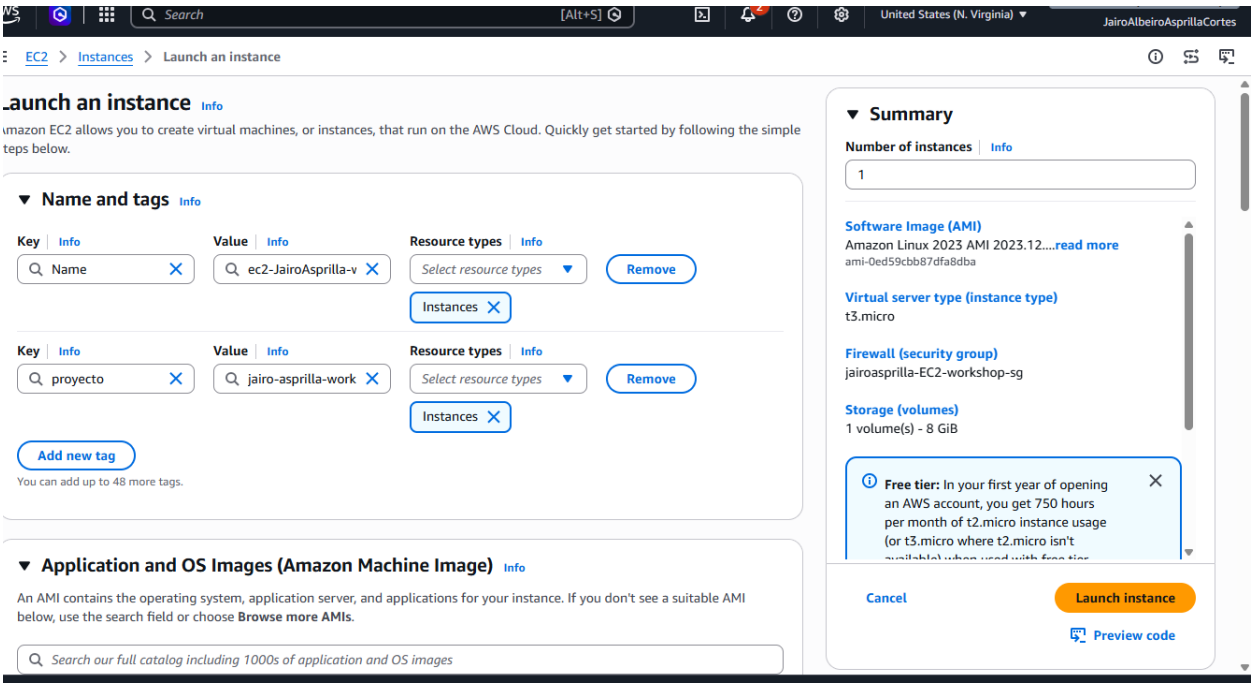


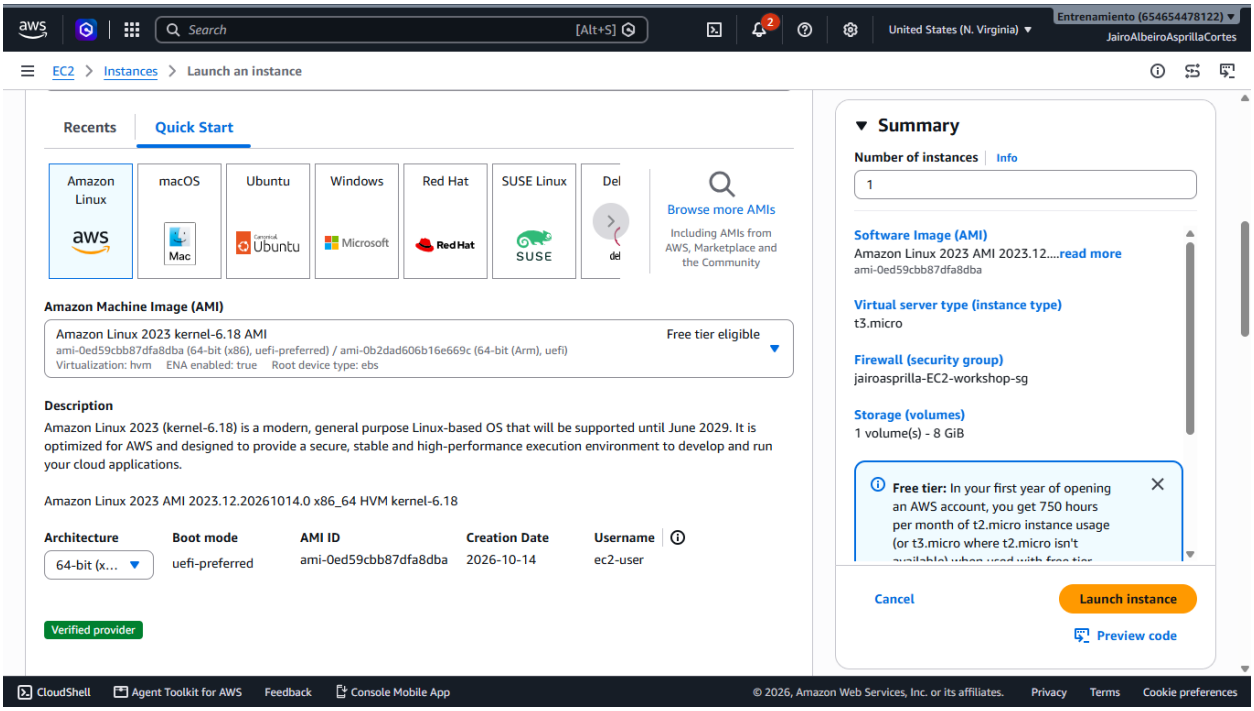

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United States (N. Virginia)
Entrenamiento (654654478122)
JairoAlbeiroAsprillaCortes
EC2 > Security Groups > Create security group
Inbound rules Info
This security group has no inbound rules.
Add rule
Outbound rules Info
Type Info
Protocol Info
Port range Info
Destination type Info
Destination Info
Description - optional Info
All traffic
All
All
Custom
0.0.0.0/0
Delete
Add rule
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags
Cancel
Create security group

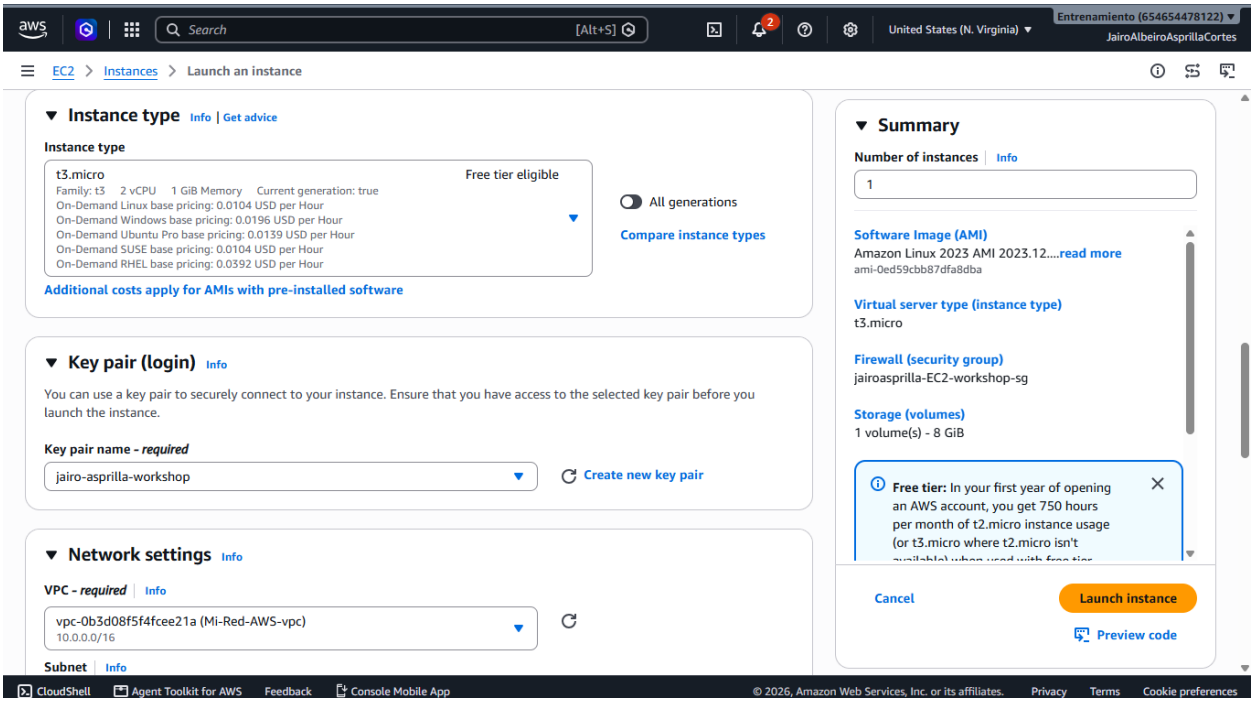

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EC2 > Security Groups > sg-0476ce19da5bef6c0
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Security group (sg-0476ce19da5bef6c0 | jairo_asprilla-rds-workshop-sg) was created successfully
Details
sg-0476ce19da5bef6c0 - jairo_asprilla-rds-workshop-sg
Actions
Details
Security group name
jairo_asprilla-rds-workshop-sg
Security group ID
sg-0476ce19da5bef6c0
Description
Permite la conexion a mi base de datos
VPC ID
vpc-0b3d08f5f4fcee21a
Owner
654654478122
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port
sgr-012b28d7efca1cd11
MYSQL/Aurora
TCP
3306

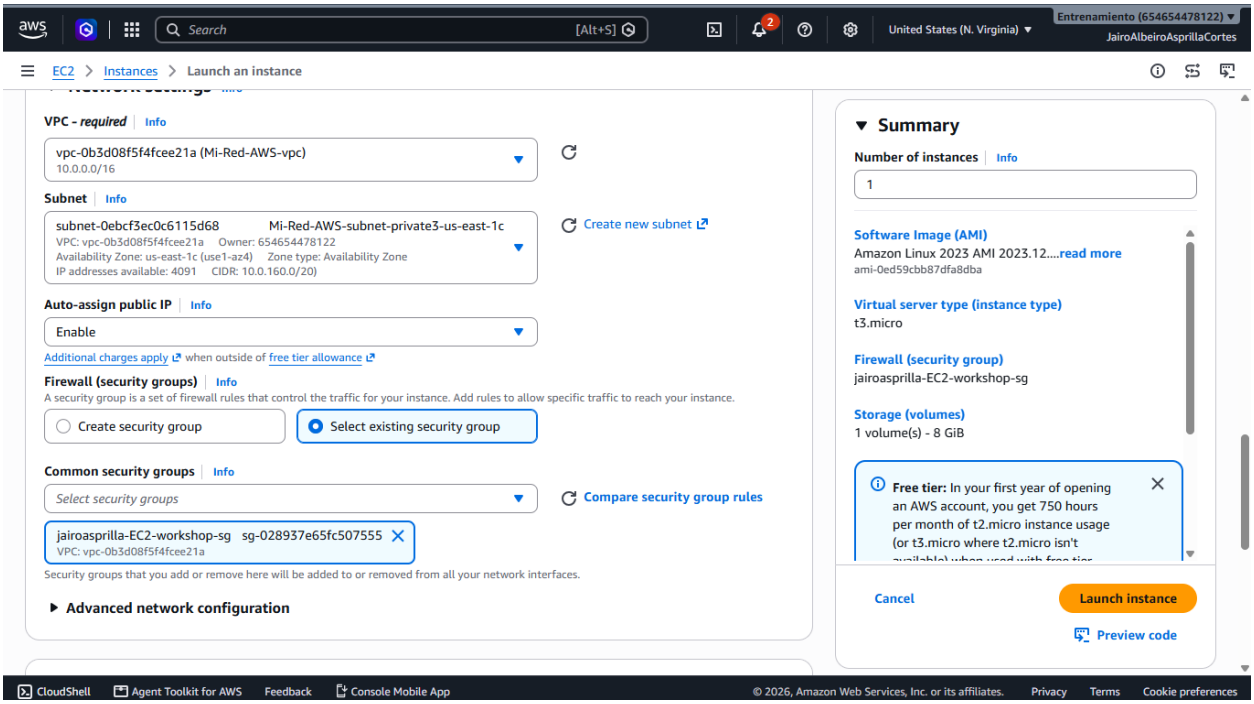


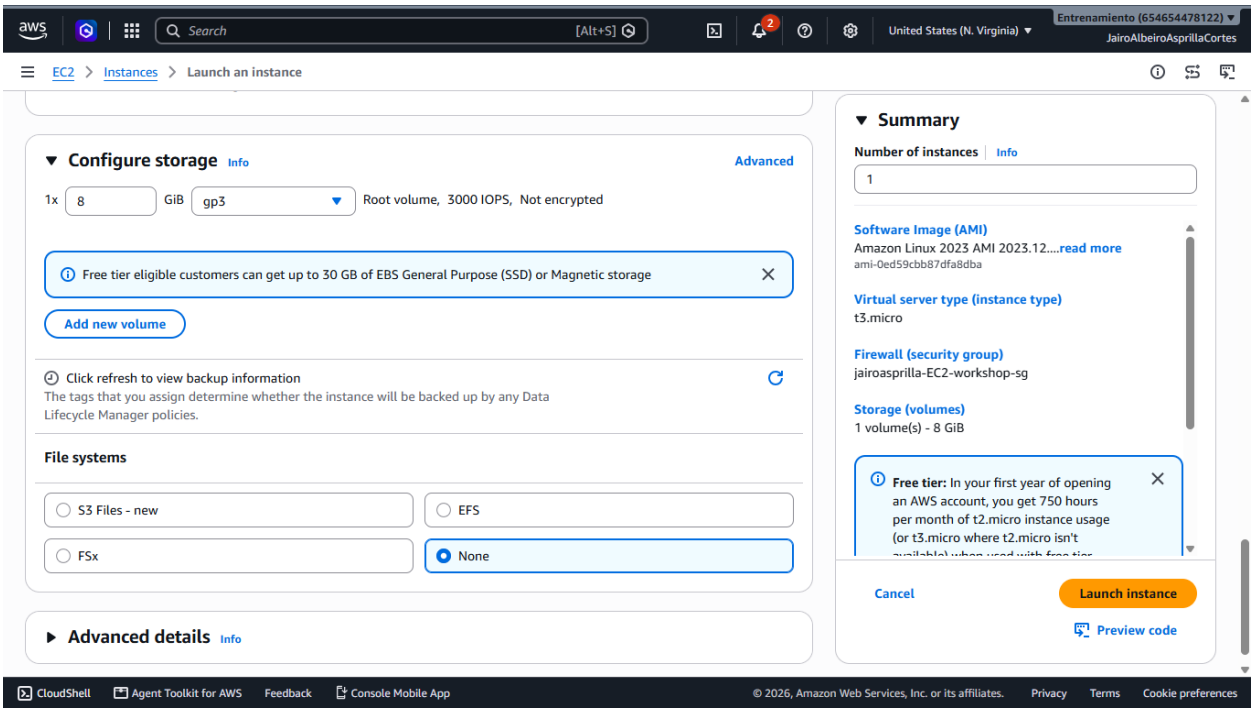
3- Crear instancia

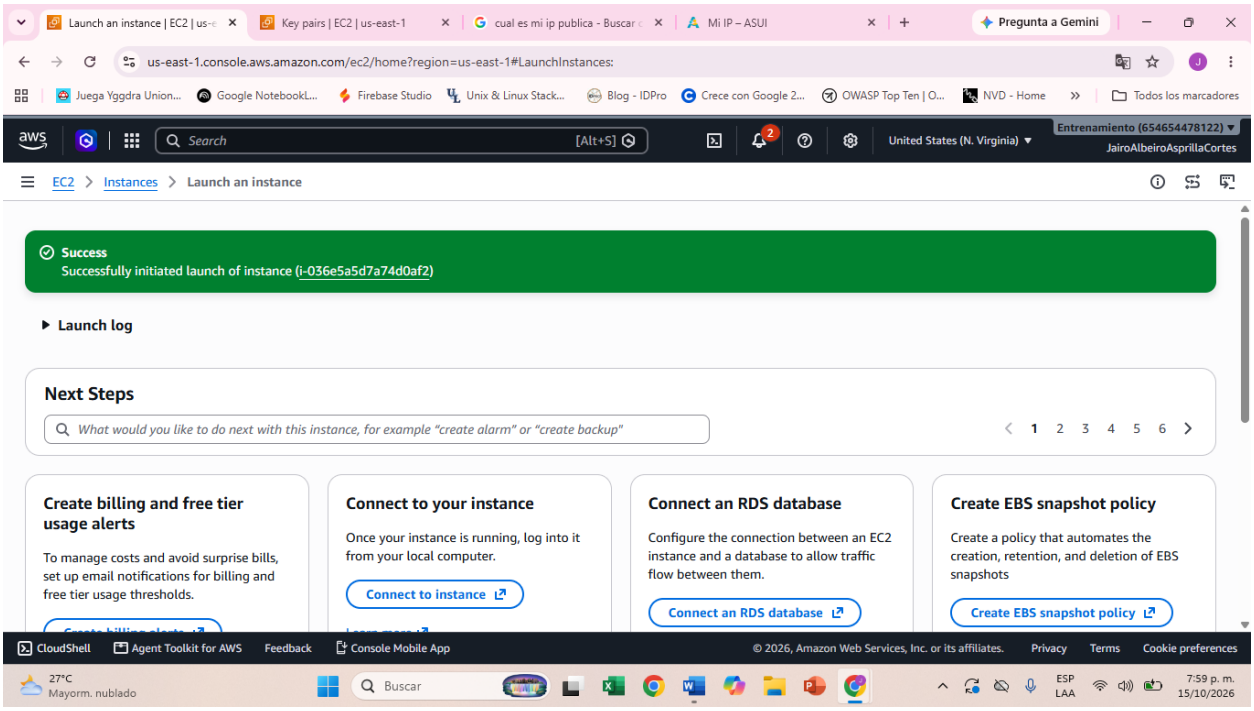



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United States (N. Virginia)
Entrenamiento (654654478122)
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EC2 > Instances > Launch an instance
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
ami-0ed59cbb87dfa8dba (64-bit (x86), uefi-preferred) / ami-0b2dad606b16e669c (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20261014.0 x86_64 HVM kernel-6.18
Architecture
Boot mode
AMI ID
Creation Date
Username
64-bit (x...
uefi-preferred
ami-0ed59cbb87dfa8dba
2026-10-14
ec2-user
Verified provider
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0ed59cbb87dfa8dba
Virtual server type (instance type)
t3.micro
Firewall (security group)
jairoasprilla-EC2-workshop-sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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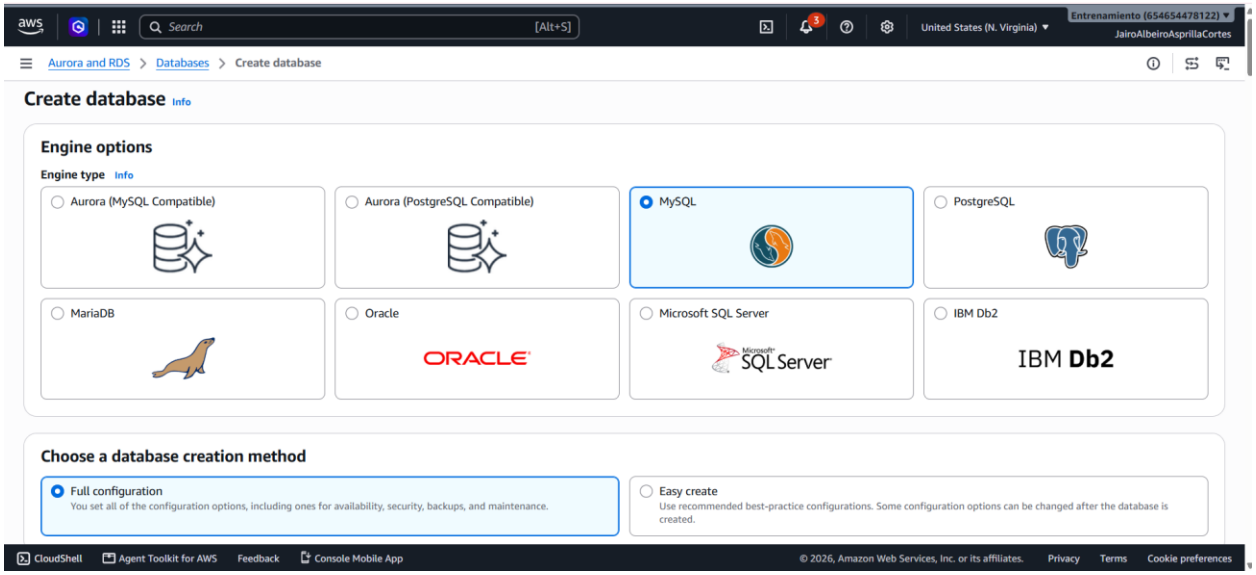

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United States (N. Virginia)
Entrenamiento (654654478122)
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EC2 > Instances > Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
jairo-asprilla-workshop
Create new key pair
Network settings Info
VPC - required Info
vpc-0b3d08f5f4fcee21a (Mi-Red-AWS-vpc)
10.0.0.0/16
Subnet Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0ed59cbb87dfa8dba
Virtual server type (instance type)
t3.micro
Firewall (security group)
jairoasprilla-EC2-workshop-sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
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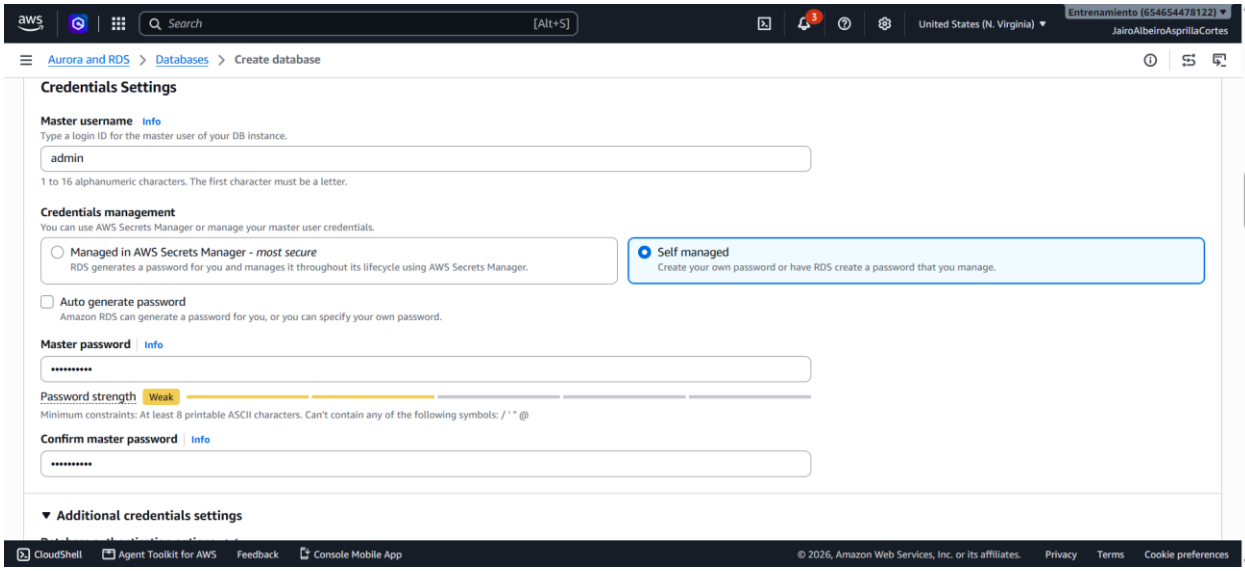

Search [Alt+S]
United States (N. Virginia)
Entrenamiento (654654478122)
JairoAlbeiroAsprillaCortes
EC2 > Instances > Launch an instance
VPC - required Info
vpc-0b3d08f5f4fcee21a (Mi-Red-AWS-vpc)
10.0.0.0/16
Subnet Info
subnet-0ebcf3ec0c6115d68 Mi-Red-AWS-subnet-private3-us-east-1c
VPC: vpc-0b3d08f5f4fcee21a Owner: 654654478122
Availability Zone: us-east-1c (use1-az4) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.160.0/20)
Create new subnet
Auto-assign public IP Info
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
jairoasprilla-EC2-workshop-sg sg-028937e65fc507555
VPC: vpc-0b3d08f5f4fcee21a
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0ed59cbb87dfa8dba
Virtual server type (instance type)
t3.micro
Firewall (security group)
jairoasprilla-EC2-workshop-sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't
Cancel
Launch instance
Preview code
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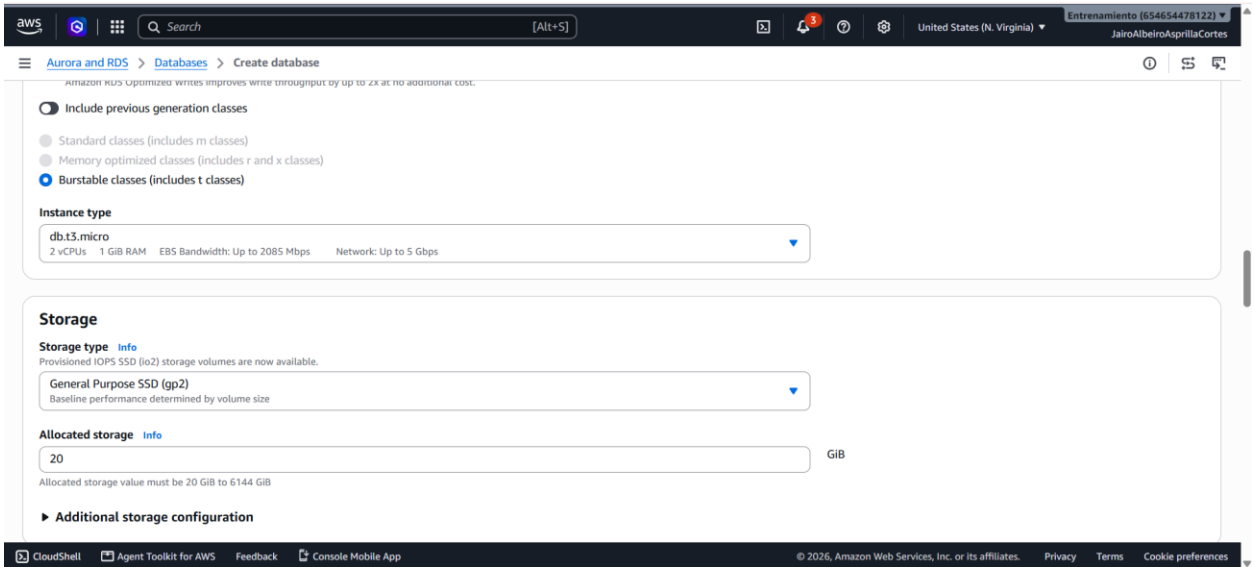

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EC2 > Instances > Launch an instance
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0ed59cbb87dfa8dba
Virtual server type (instance type)
t3.micro
Firewall (security group)
jairoasprilla-EC2-workshop-sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't
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Launch instance
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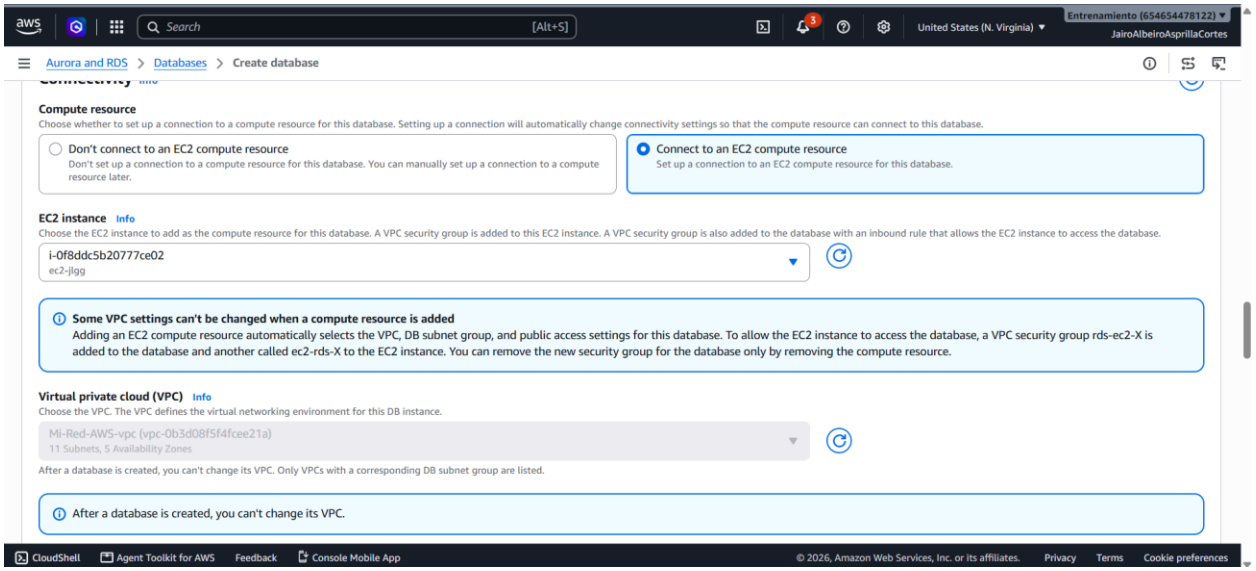


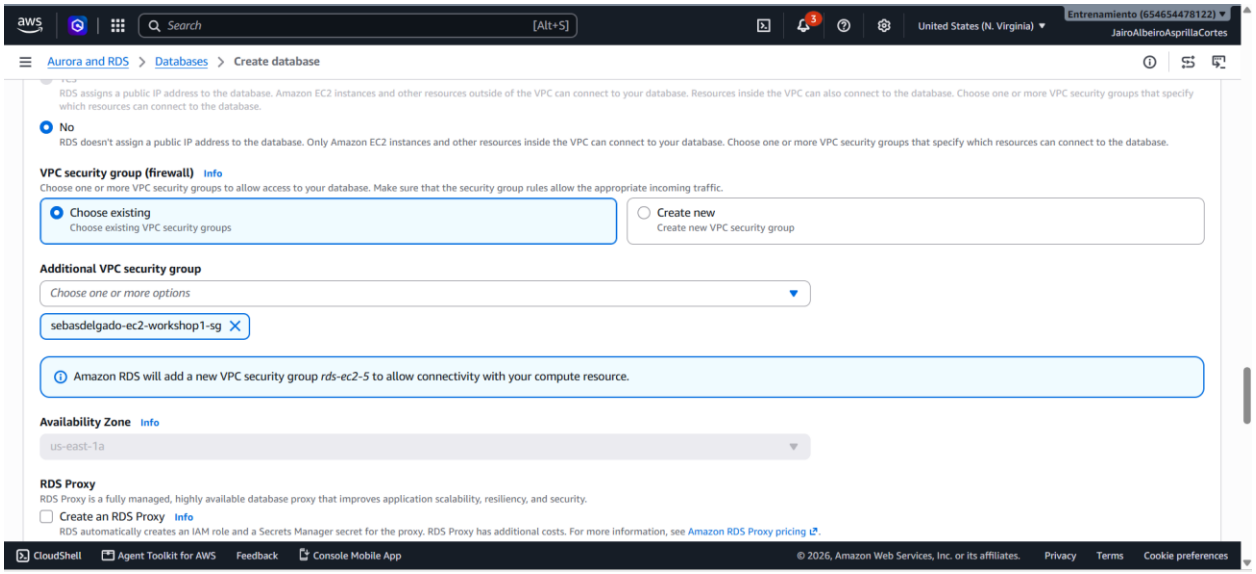
4- Creación de nuestra base de datos MySql



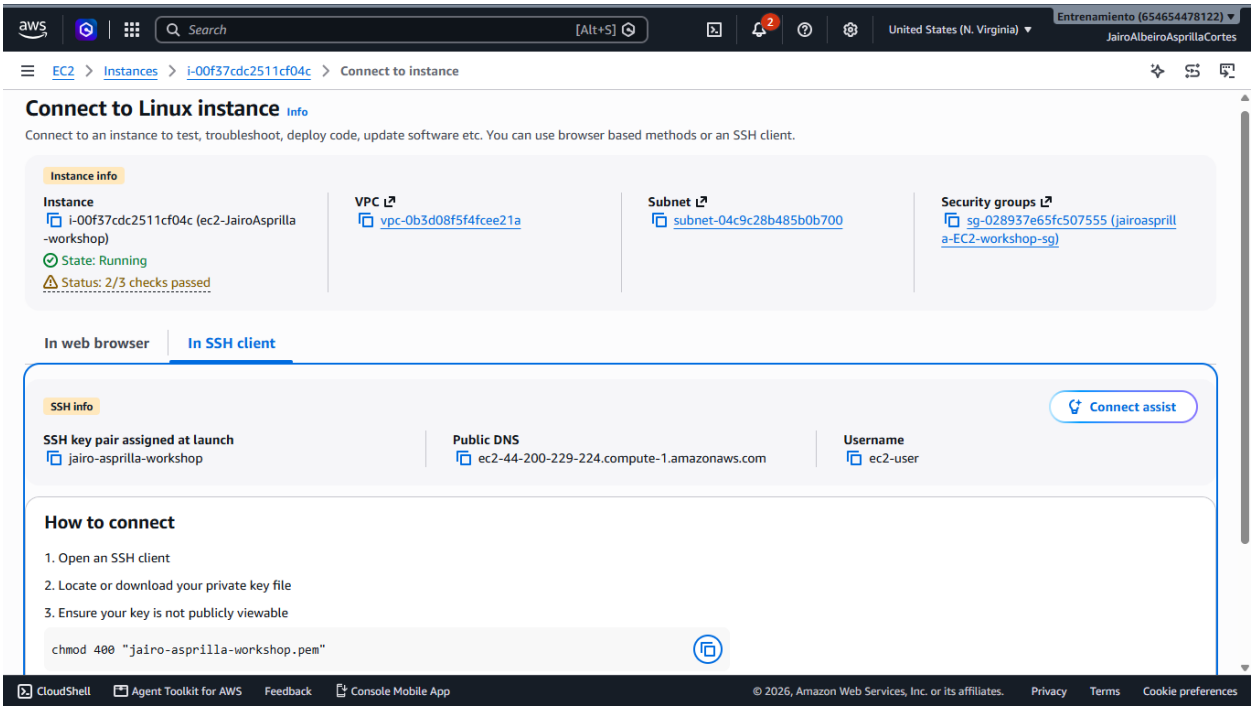

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Aurora and RDS
Databases
Create database
Credentials Settings
Master username Info
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.
Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
Self managed
Create your own password or have RDS create a password that you manage.
Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.
Master password Info
Password strength
Weak
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @
Confirm master password Info
Additional credentials settings
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Agent Toolkit for AWS
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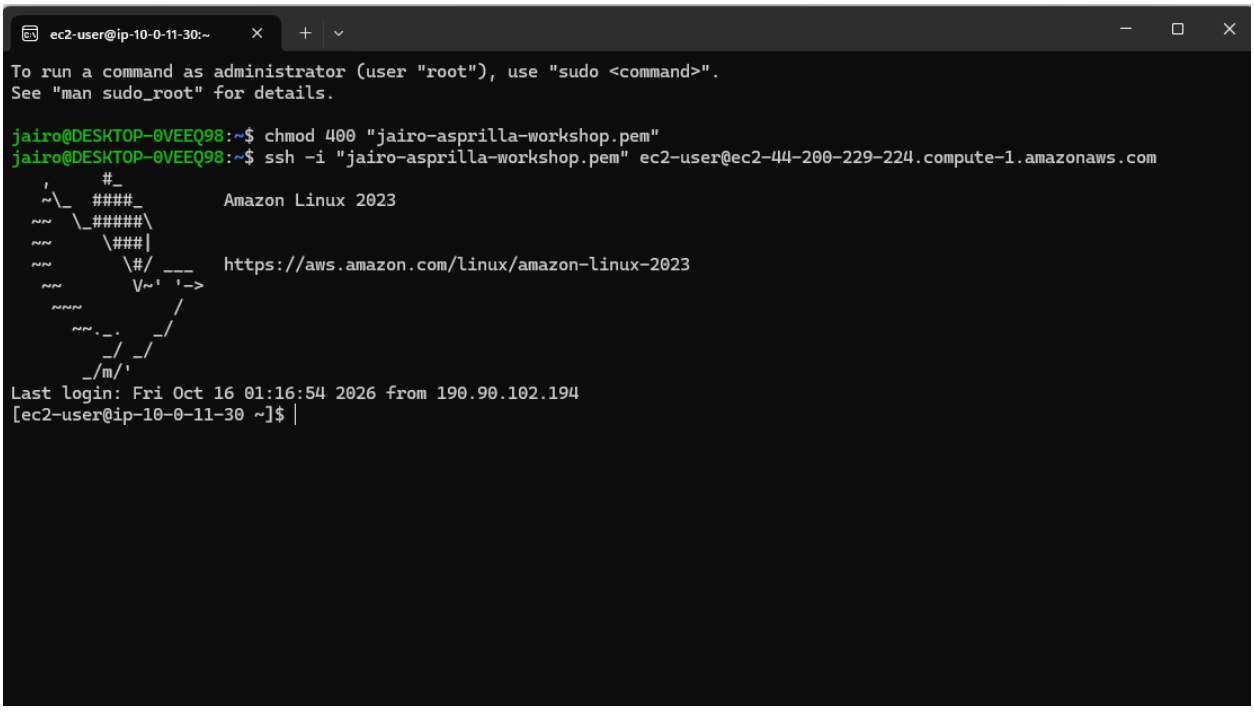

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Aurora and RDS
Databases
Create database
Include previous generation classes
Standard classes (includes m classes)
Memory optimized classes (includes r and x classes)
Burstable classes (includes t classes)
Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2085 Mbps Network: Up to 5 Gbps
Storage
Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size
Allocated storage Info
20
GiB
Allocated storage value must be 20 GiB to 6144 GiB
Additional storage configuration
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Aurora and RDS > Databases > Create database
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
EC2 instance Info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.
i-0f8ddc5b20777ce02
ec2-jlgg
Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
Mi-Red-AWS-vpc (vpc-0b3d08f5f4fcee21a)
11 Subnets, 5 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
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United States (N. Virginia)
Entrenamiento (654654478122)
JairoAlbeiroAsprillaCortes
Aurora and RDS > Databases > Create database
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Additional VPC security group
Choose one or more options
sebasdelgado-ec2-workshop1-sg
Amazon RDS will add a new VPC security group rds-ec2-5 to allow connectivity with your compute resource.
Availability Zone Info
us-east-1a
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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5- Conectarnos a la instancia vía SSH





6- Instalar MariaDB

```
ec2-user@ip-10-0-11-30:~
        _/ _/
      _/m/'
Last login: Fri Oct 16 01:17:47 2026 from 190.90.102.194
[ec2-user@ip-10-0-11-30 ~]$ sudo dnf install mariadb105
Amazon Linux 2023 Kernel Livepatch repository                    707 kB/s |  89 kB     00:00
Last metadata expiration check: 0:00:01 ago on Fri Oct 16 02:24:49 2026.
Dependencies resolved.
================================================================================================================
 Package                        Architecture     Version                        Repository             Size
================================================================================================================
Installing:
 mariadb105                     x86_64           3:10.5.29-1.amzn2023.0.1       amazonlinux           1.5 M
Installing dependencies:
 mariadb-connector-c            x86_64           3.3.10-1.amzn2023.0.2          amazonlinux           211 k
 mariadb-connector-c-config     noarch           3.3.10-1.amzn2023.0.2          amazonlinux           9.3 k
 mariadb105-common              x86_64           3:10.5.29-1.amzn2023.0.1       amazonlinux            28 k
 perl-Sys-Hostname              x86_64           1.23-477.amzn2023.0.10         amazonlinux            17 k

Transaction Summary
================================================================================================================
Install  5 Packages

Total download size: 1.8 M
Installed size: 19 M
Is this ok [y/N]: y
Downloading Packages:
(1/5): mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64.rpm      5.9 MB/s | 211 kB     00:00
(2/5): mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch.rpm 250 kB/s | 9.3 kB     00:00
(3/5): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm       895 kB/s |  28 kB     00:00
(4/5): perl-Sys-Hostname-1.23-477.amzn2023.0.10.x86_64.rpm       518 kB/s |  17 kB     00:00
```

```
ec2-user@ip-10-0-11-30:~
(3/5): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm       895 kB/s |  28 kB     00:00
(4/5): perl-Sys-Hostname-1.23-477.amzn2023.0.10.x86_64.rpm       518 kB/s |  17 kB     00:00
(5/5): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm              8.6 MB/s | 1.5 MB     00:00
----------------------------------------------------------------------------------------------------------------
Total                                                            6.0 MB/s | 1.8 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                              1/1
  Installing       : mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch                      1/5
  Installing       : mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64                             2/5
  Installing       : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64                            3/5
  Installing       : perl-Sys-Hostname-1.23-477.amzn2023.0.10.x86_64                              4/5
  Installing       : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64                                   5/5
  Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64                                   5/5
  Verifying        : mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64                             1/5
  Verifying        : mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch                      2/5
  Verifying        : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64                                   3/5
  Verifying        : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64                            4/5
  Verifying        : perl-Sys-Hostname-1.23-477.amzn2023.0.10.x86_64                              5/5

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64        mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64              mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  perl-Sys-Hostname-1.23-477.amzn2023.0.10.x86_64

Complete!
[ec2-user@ip-10-0-11-30 ~]$
```

7- Conectarnos a nuestra RDS desde nuestra instancia

